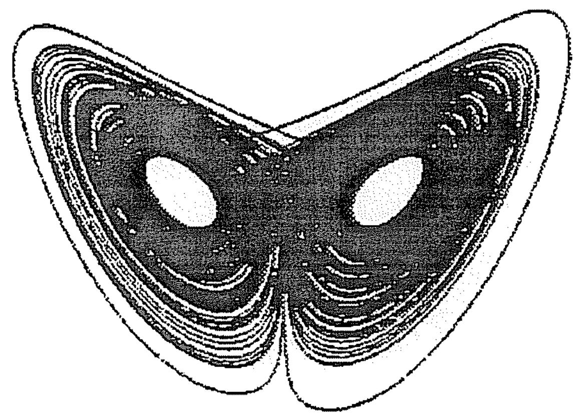
Nonlinear Science Today



Projection of the strange attractor on the field-inversion plane [See Fig. 2(b)]. Reprinted from Hübner et al., Phys. Rev. A 40, 6354–6365 (1989) (See Ref. [2]).

The Dynamics of Optical Systems: A Renaissance of the 1990s

*Daniel J. Gauthier**

Why do some lasers produce noise-like intensity spikes, and what is the optimum implementation of an optical switch? Such questions have motivated researchers over the last three decades to uncover the sources of instability in nonlinear optical systems. Recently, there has been a surge in the number of researchers investigating the dynamics of optical systems, because it now seems possible to tackle problems such as optical turbulence and controlling optical chaos that seemed intractable a decade ago. I will briefly trace the historical development of our understanding of the dynamics of optical systems and highlight some of the recent trends.

I. Introduction

Many of the researchers using lasers have been confronted by the appearance of ''noise-like'' intensity fluctuations or ''turbulent-like'' spatial patterns in the laser's beam. This type of behavior was clearly evident even during the earliest investigations of lasers in the 1960s [1,2] where it was found that the intensity of the light generated by the ruby laser displayed irregular spiking, as shown in Fig. 1. Was the spiking due to inadequate shielding of the laser from the environment or was is it due to some intrinsic property of the laser?

After many years of research on nonlinear optical systems we now understand that these observed instabilities can arise from deterministic forces that govern the interplay between the radiation field and matter. The path to this realization was rather long because the origin of the instabilities seemed to be very complicated, and it took several years to place the research in the framework of universal behavior of dynamics systems. In hindsight, it is not surprising that optical systems readily display instabilities because the intense, coherent radiation generated by lasers can easily induce a nonlinear response in matter.

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*Department of Physics and Center for Complex and Nonlinear Systems, Duke University, Durham, NC 27708.

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Given the ubiquitous nature of optical instabilities, researchers have been striving for the past thirty years to understand their origin so that they can be fully exploited, avoided, or controlled. Significant progress has been made to the point where we have developed a rather complete understanding of the dynamical behavior for some relatively simple optical systems [3–25]. For example, some classes of laser instabilities [21] and optical bistability [9] are well understood.

The motivation for studying optical instabilities is multifaceted and depends on the background of the researcher. For some, instabilities are viewed as a limitation to the performance of the optical system that must be avoided or controlled. For example, optical engineers try to quell chaotic fluctuations in the intensity of diode lasers because they limit the ability to detect information stored on compact disks [28–30]. In contrast, other researchers have put the unstable behavior to good use making practical devices such as optical switches (based on optical bistability [9]). Yet other researchers investigate unstable optical systems as a paradigm for complexity.

As our understanding of the dynamics of simple optical systems matures, many researchers armed with a collection of new theoretical and experimental tools [27] are beginning to examine previously investigated systems as well as entirely new classes of complex optical systems. In my opinion, a renaissance began in 1990 when there was a major shift in research toward the investigation of spatially extended optical systems. The shift was made obvious by the large collection of papers that were published on this subject in two special issues of the *Journal of the Optical Society of America* [27]. The renaissance was further invigorated by a breakthrough in the nonlinear dynamics community for controlling chaos in low-dimensional systems [31–33]. The control scheme was used recently to stabilize chaos in lasers [34–41], which is an important achievement from a practical as well as fundamental perspective.

I will briefly review some of the research prior to 1990 that laid the foundation for a surge of interest in the dynamics of optical systems, and I will describe some of the current research trends on spatially extended systems and on controlling optical instabilities.

II. The Dynamics of Single-Transverse-Mode Optical Systems

Conceptually, it is quite easy to describe the dynamical evolution of an optical system: the electromagnetic field polarizes the medium (as prescribed by quantum mechanics, for example) which in turn acts as a source of new fields (as prescribed by Maxwell's equations, for example). While it is possible to write down a complete model using these concepts [42–51], it is not always practical, because the dynamics are governed by a complex interplay between several different nonlinear optical processes, diffraction and boundary conditions. For this reason most of the early research was devoted to identifying the simplest form of the wave equation and the nonlinear polarization that would qualitatively describe the observed instabilities. Somewhat surprisingly at the time, it was found that very simple models of the interaction could display instabilities if they incorporated the effects of gain, nonlinearity and, for the case of absolute instabilities, some form of optical feedback. In the next section I will describe one model that was used to investigate the origin of the spiking behavior observed in lasers.

One feature of optical systems that greatly simplifies their analysis is that the number of excited modes tends to be incredibly small. Even though the physical scale of a system is of the order of 10^5 optical wavelengths, only a narrow band of wavelengths ($\Delta\lambda/\lambda \sim 10^{-5}$) is excited because the gain is sharply peaked, and only a small number of transverse wave vectors is excited because of boundary conditions in the system, such as apertures. Hence, the time- (spatial-) scale of temporal instabilities (patterns) tend to be much slower (larger) than the inverse of the optical frequency (wavelength) since they arise from beating of the excited modes. Researchers took advantage of this feature to reduce significantly the complexity of the models.

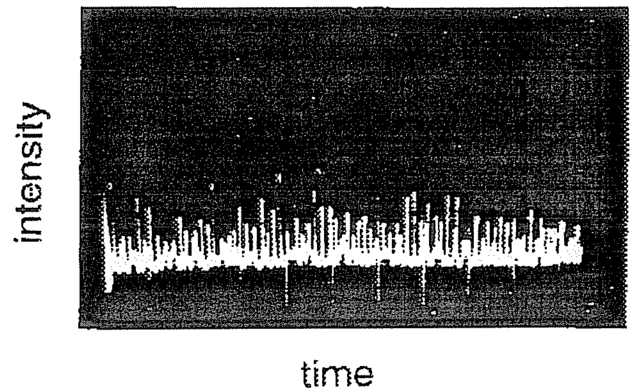

Figure 1. Temporal evolution of the intensity emitted by a ruby laser. Reprinted from D. F. Nelson and W. S. Boyle, A continuously operating ruby optical laser, Appl. Opt. **1**, 181–183 (1962).

Many of the early models of optical systems assumed that the spatial extent of the system transverse to the propagation direction was limited because it was intractable to solve for the dynamics of spatially extended systems except in the simplest cases [52–54] and it was possible to configure an experiment to operate in this regime. The models based on this approach were quite successful in that they lead to an intuitive understanding of the origin of instabilities in optical systems, and they predicted qualitatively the occurrence of instabilities. The major shortcoming of some of the models was that the predicted threshold for instabilities was much higher than that observed experimentally. This problem could be corrected to some extent by accounting for the transverse profile of the laser beams [55–56]. Motivated by these observations, researchers began to develop new theoretical and computational techniques to more accurately model the transverse nature of the field. In my opinion, the modern era of research on the dynamics of optical systems began with the development of these new techniques that allow us to routinely investigate spatially extended systems.

Transverse effects are important when the aspect ratio, which is quantified by a dimensionless parameter F called the Fresnel number [57], of the systems becomes large. It is defined through the relation

$$F = \frac{a^2}{\lambda L}, \tag{1}$$

where a is the transverse dimension, λ is the wavelength of the optical radiation, and L is the longitudinal dimension (typically the length of the resonator or the length of the medium). When $F \leq 1$, the transverse spatial profile is usually stationary and is well characterized by a single transverse-mode of the field. In contrast, complex spatial patterns can develop in extended systems ($F >> 1$) where the spread of the beam due to diffraction is smaller than the geometric acceptance angle of the system. Under special circumstances the spatial extent can be adjusted smoothly from low- to high-dimensional, which offers a unique advantage to study its effects on the dynamics of the system.

III. The Single-Mode Laser Model

One of the earliest models used to investigate the origin of the spiking behavior observed in lasers is known as the single-mode laser model, which describes in a self-consistent manner the interaction between the field circulating in the laser resonator and the gain medium [58–61]. For simplicity, it is assumed that the gain medium consists of a collection of homogeneously broadened two-level atoms (polarization p and inversion w) and that the field is well described by a single

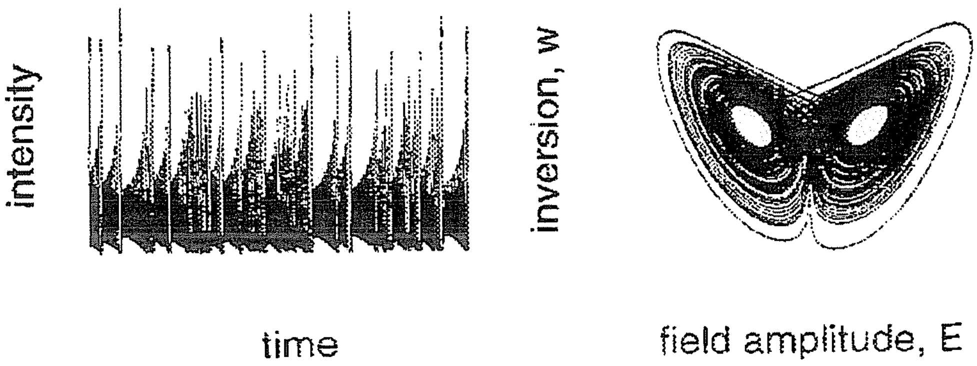

Figure 2. Predictions of the single-mode-laser model. (a) Temporal evolution of the laser intensity. (b) Projection of the strange attractor on the field-inversion plane. Reprinted from U. Hübner, N. B. Abraham, and C. O. Weiss, Dimensions and entropies of chaotic intensity pulsations in a single-mode far-infrared NH_3 laser, Phys. Rev. A **40**, 6354–6365 (1989).

plane-wave mode of the field (amplitude E) whose frequency is equal to the atomic transition frequency. In addition, it is assumed that the resonator has low loss so that the spatial variation of the field along the axis of the resonator can be ignored (the mean-field approximation). Note that the mode selectivity of the high-finesse cavity removes effectively any spatial variation of the field. Under these conditions the dynamical evolution of the laser is given by

$$\dot{E} = -\kappa\,(E - p),$$
$$\dot{p} = -\gamma_\perp\,(p - Ew), \qquad (2)$$
$$\dot{w} = -\gamma_\parallel\,(w + Ev - w_p),$$

where $\kappa, \gamma_\perp, \gamma_\parallel$ denote the decay rate of the field in the cavity, the atomic polarization, and inversion, respectively, and w_p is the equilibrium inversion due to the pump process. The laser has the necessary ingredients to display instabilities: gain arises from stimulated emission, nonlinearity appears due to saturation of the inversion, and feedback is provided by the resonator.

The first laser threshold occurs when the gain experienced by the field in a round-trip through the resonator is equal to the loss ($w_p = 1$), that is, the laser turns on. For a "good" cavity ($\kappa < \gamma_\perp + \gamma_\parallel$) the model predicts that the field is always steady for arbitrary pump rates, which is desirable for many applications. In contrast, the field is unstable for a "bad" cavity above the second laser threshold that is defined by

$$w_p^{th} = \frac{\kappa(\gamma_\parallel + \kappa + 3\gamma_\perp)}{(\kappa - \gamma_\parallel - \gamma_\perp)}\,. \qquad (3)$$

From Eq. (3) it is easy to see that the minimum threshold for instabilities occurs at a pump rate nine times greater than the first laser threshold ($w_p = 9$). Typically, it is found that the intensity of the laser (proportional to E^2) displays wild temporal fluctuations above the second laser threshold as shown in Fig. 2a, which is for the case $\gamma_\parallel = 0.25\gamma_\perp$, $\kappa = 2\gamma_\perp$, and $w_p = 15$. This behavior, termed "autostochastic" by early researchers, is reminiscent of the noise-like behavior observed in the ruby laser (Fig. 1). Unfortunately, Eq. (2) does not faithfully represent the dynamics of the ruby laser because the instabilities observed in the experiments occurred just above the first laser threshold.

Even though there were discrepancies between the predictions of the model and observations, many researchers were intrigued by appearance of "noise-like" behavior from a simple set of differential equations. The behavior becomes even more interesting when we view the dynamics in phase space as shown in Fig. 2b, which is a projection of the trajectory on the $E - w$ plane for the same conditions used to generate Fig. 2a. Surprisingly, the strange attractor looks very similar to the now-familiar Lorenz attractor that characterizes hydrodynamic convection. In fact, Haken [62] showed in 1975 that the single-mode laser-model is isomorphic with the Lorenz model [63], clarifying the similarity between models of instabilities in optical systems and in other disciplines.

How strong is the analogy between optics and hydrodynamics? To answer this question, researchers expended considerable effort to find a laser system that was well represented by Eq. (2). The search was impeded by the opposing requirements of a bad cavity operating far above the first laser threshold, but it was felt that the conditions could be fulfilled using a class of laser-pumped far-infrared lasers [64]. Although the observed instability threshold corresponded closely to that predicted by the model, there was some controversy concerning the measurements because they could not distinguish between motion in phase space about a single unstable fixed point and motion about two unstable fixed points as shown in Fig. 2b. This ambiguity arose because the intensity of the light emitted by the laser was recorded rather than the electric field amplitude. To quell the controversy, Weiss and co-workers [65,66] set up a laser-heterodyne detector that could measure the field amplitude. They observed that the field experienced abrupt π phase changes as shown in Fig. 3, which corresponds to the trajectory switching from one spiral to the other. In addition, they found that the dimensions and entropies of the attractor reconstructed from experimental data coincided with those of the attractor shown in Fig. 2b within the experimental accuracy. Hence, it appears that the Lorenz–Haken model adequately characterizes the dynamics of the far-infrared laser.

Quite different behavior was observed [65–68] in the laser when its frequency was detuned from the atomic transition frequency, however. New routes to chaos were identified that had not been observed previously, and it was found that homoclinic chaos could exist in the system. These results point out that the dynamics of optical systems are indeed rich and complement the studies of hydrodynamic systems rather than merely providing a "dry" technique for performing the experiments.

IV. Intrinsic Optical Instabilities

The success enjoyed by researchers studying the far-infrared laser was also experienced by researchers investigating different single-mode systems. For example, our understanding of instabilities in multi-frequency lasers [21] and optical bistability [69–75] matured during this period. Interestingly, all of the systems that displayed chaotic instabilities contained optical feedback provided by mirrors external to the nonlinear optical medium. Hence, it was natural to presume that

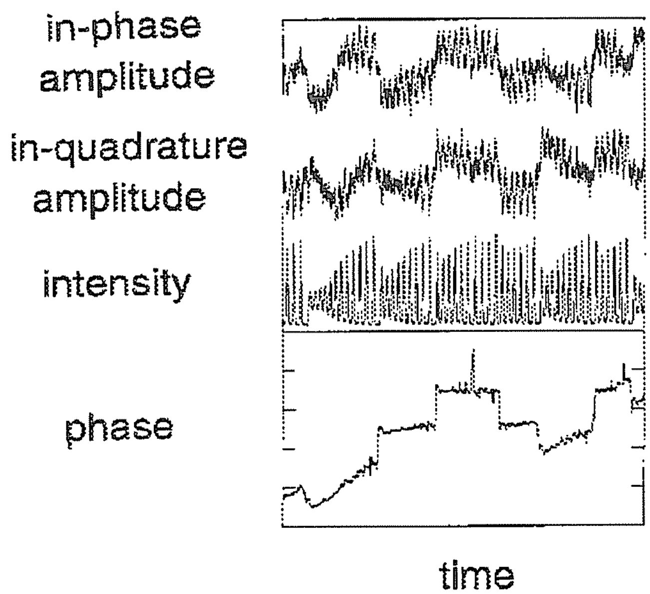

Figure 3. Chaotic pulsing of far-infrared laser emission. The π jumps in phase (one division on the verticle axis) indicate when the trajectory switches from one spiral arm of the attractor to the other. Reprinted from C. O. Weiss, N. B. Abraham, and U. Hübner, Homoclinic and heteroclinic chaos in a single-mode laser, Phys. Rev. Lett. **61**, 1587–1590 (1988).

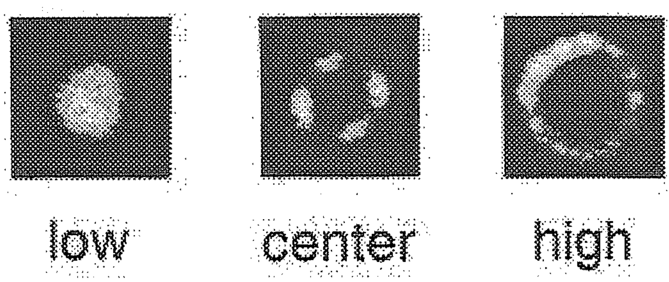

Figure 4. The radiation patterns of the light emitted in the orthogonal-polarization component for laser beams counter-propagating in sodium vapor when the laser is tuned to the low-, center, and high-side of the atomic resonance frequency. Reprinted from D. J. Gauthier, M. S. Malcuit, and R. W. Boyd, Polarization instabilities of counterpropagating laser beams in sodium vapor, Phys. Rev. Lett. **61**, 1827–1830 (1988).

external feedback was necessary to observe instabilities.

Silberberg and Bar-Joseph [76,77] discovered that it was possible to observe instabilities in laser beams counterpropagating through a nonlinear medium *without* external mirrors while they were investigating the origin of instabilities in nonlinear ring cavities. They found that feedback can be provided directly by the nonlinear interaction due to a distributed feedback structure that is formed by the interference of the waves. As the intensity of the laser beams is increased from low to high, the intensity of the transmitted beams undergo a period-doubling route to chaos due to the combined action of nonlinear gain and distributed feedback. Preliminary experiments performed by Khitrova and co-workers [78] verified that laser beams counterpropagating through a sodium vapor are unstable to the growth of new beams that fluctuate periodically.

Several other researchers showed that the threshold for instabilities can be reduced significantly when transverse effects [79,80] and the vector nature of the field [81] are incorporated into the theoretical formulation. These predictions were verified by two groups that also investigated the stability of beams counterpropagating through a sodium vapor. Grynberg and co-workers [82–84] found that the radiation arising from the instability was emitted in a cone of light surrounding the transmitted laser beams and that under some conditions the cone fragments into a hexagonal pattern of beams. The hexagonal pattern is intriguing because similar patterns are observed in many other pattern-forming systems in nature. For the present case, the pattern can be explained by considering the conservation of momentum (phase-matching) in the interaction.

Gauthier and co-workers [85,86] observed that the polarization states of the transmitted beams could be unstable while the total intensity of the beam remained stable. The behavior was dependent on the detuning of the laser from the atomic transition frequency because the tensor nature of the interaction is frequency dependent. They found that the light emitted from the medium orthogonal to the input polarization (power $P_\perp$) could be in the form of nonuniform patterns as shown in Fig. 4 and that $P_\perp$ displayed multistability (see Fig. 5) and chaotic instabilities (see Fig. 6) for increasing nonlinearity.

These results demonstrate that an incredibly simple system [87–90] is capable of behaving in a very complex manner. One advantage of this system in comparison to other optical systems that contain external mirrors is that there are fewer boundary conditions that limit the number of modes that can be excited by an instability. Future research on this system will have to address whether complex spatial–temporal patterns develop when the Fresnel number of the interaction is increased.

Taking the simplifying approach even further, researchers have found that a *single* beam of light propagating through a nonlinear medium can display complex spatial–temporal behavior when an appropriate spatial pattern is imposed on the input beam. Grantham *et al.* [91] and Law and Swartzlander [92] observed the formation of transverse solitary waves, optical vortices, and spatial–temporal instabilities. I am sure that future research on this system will greatly enhance our understanding of the dynamics of optical systems because of its simplicity.

V. Complex Spatial Patterns in Optical Systems

Currently, many researchers are investigating the dynamics of self-organized, complex spatial patterns in optical systems [93–97]. Not surprisingly, we find that most systems that display dynamical instabilities when the spatial extent of the system is limited also display spontaneous pattern formation and spatial–temporal instabilities. The patterns observed in optics are very similar to those observed in fluids and other systems, which may seem surprising because the systems are so different. For example, diffusion in optical systems arises from diffraction of the optical field rather than the motion of particles as in hydrodynamics. Most certainly, the patterns are similar because the mathematical formulation is similar, and the patterns are not overly sensitive to the explicit form of the nonlinearity. It also attests to an underlying (complex) symmetry of nature that we are trying to understand.

The research on extended systems is fast-paced and covers a broad range of topics [24,25]. To give a sense for some of the current topics, I will highlight the behavior of a few systems that are under current investigation. Feng *et al.* [98] have reinvestigated the dynamics of the single-mode laser discussed in Sec. III with (one-dimensional) transverse effects incorporated into the formalism. They found that the laser displays travelling waves in the transverse section of the beam as shown in Fig. 7 and that the existence of these patterns lowers the threshold for lasing. Complex patterns also occur in "passive" optical systems (no energy

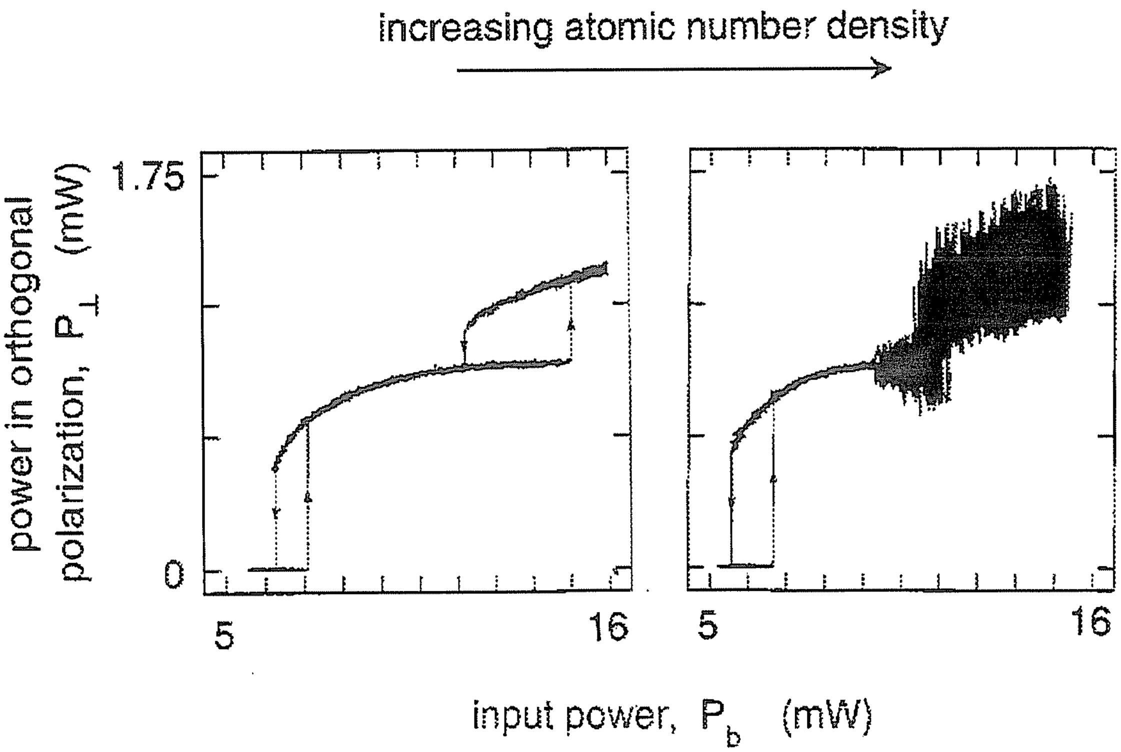

Figure 5. Bistability and instabilities in the polarization of laser beams counterpropagating through sodium vapor. Power emitted in the forward direction in the polarization component orthogonal to that of the input waves for increasing atomic number density. Reprinted from D. J. Gauthier, M. S. Malcuit, and R. W. Boyd, Polarization instabilities of counterpropagating laser beams in sodium vapor, Phys. Rev. Lett. **61**, 1827–1830 (1988).

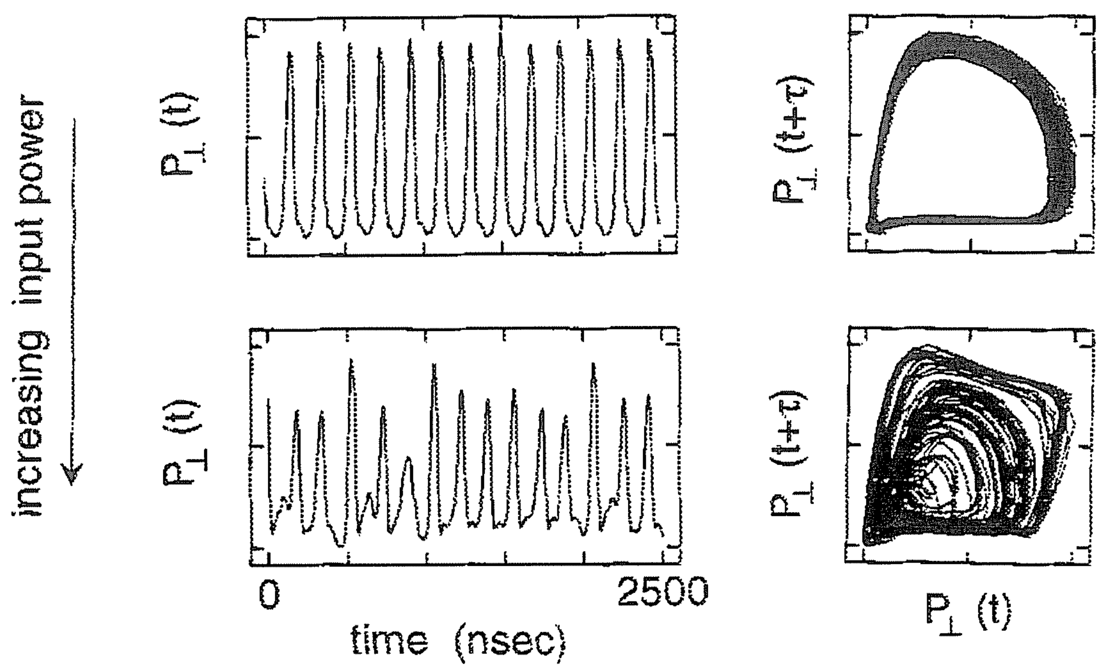

Figure 6. Periodic and chaotic instabilities in the polarization of laser beams counterpropagating through sodium vapor. Reprinted from D. J. Gauthier, M. S. Malcuit, A. L. Gaeta, and R. W. Boyd, Polarization bistability of counterpropagating laser beams, Phys. Rev. Lett. **64**, 1721–1724 (1990).

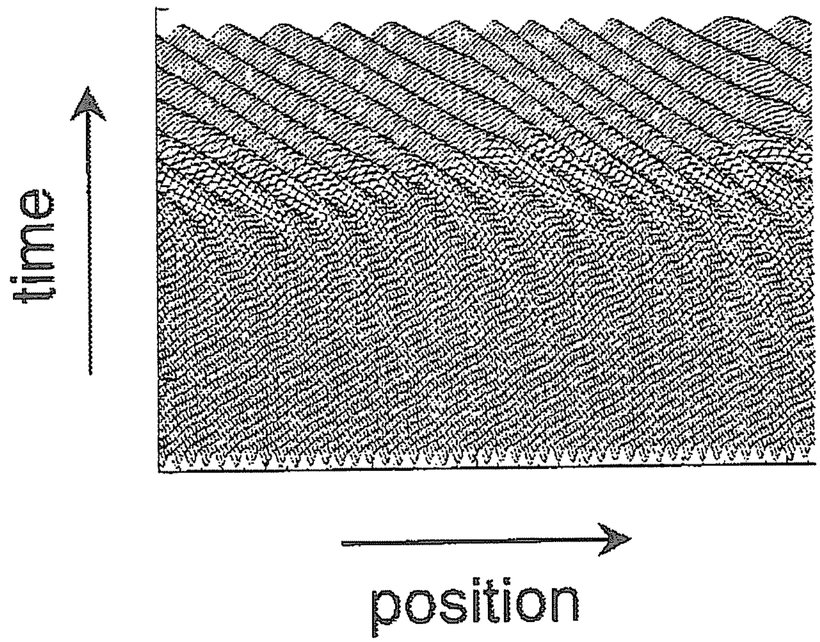

Figure 7. Transition from right-travelling wave to left-travelling wave in the spatial profile of a single-mode laser. Reprinted from Q. Feng, J. V. Moloney, and A. C. Newell, Amplitude instabilities of transverse traveling waves in lasers, Phys. Rev. Lett. **71**, 1705–1708 (1993).

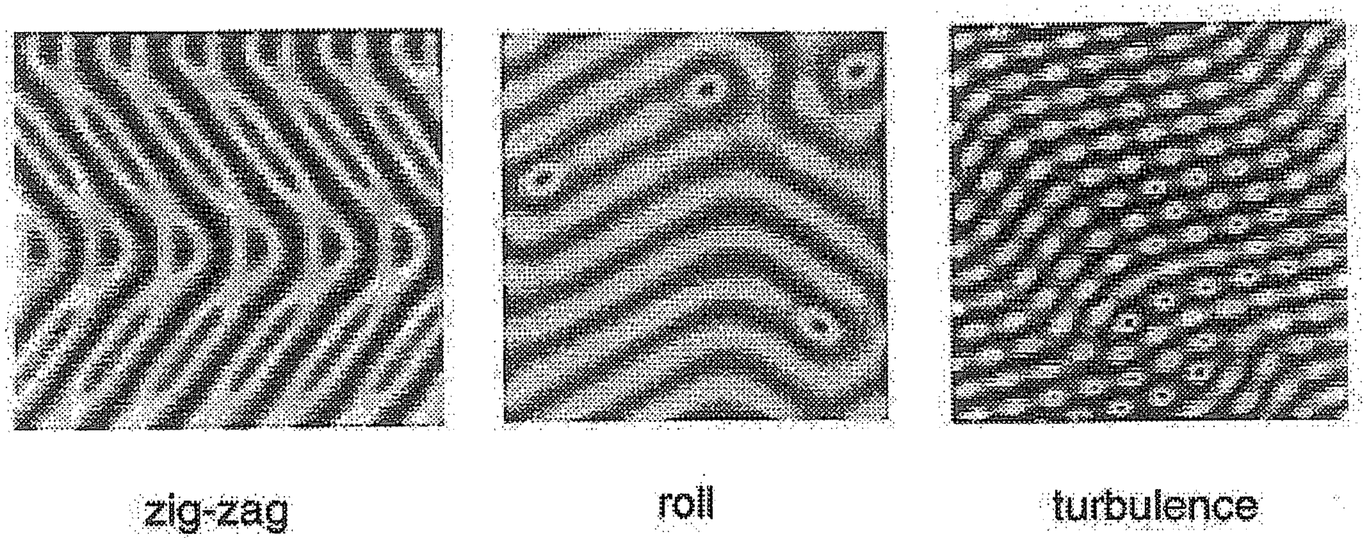

Figure 8. Spatial patterns in the radiation field emitted by an optical parametric oscillator. Reprinted from M. Brambilla, D. Camesasca, A. Gatti, L. A. Lugiato, and G. L. Oppo, Spontaneous spatial structures and spatial squeezing in optical parametric oscillators, in *Chaos in Optics*, R. Roy, Ed., Proc. Soc. Photo-Opt. Instrum. Eng. **2039** (1993), pp. 282–289.

pumped into the system from an external source) such as optical parametric amplifiers, as shown in Fig. 8 [99,100], and a laser-driven nonlinear medium in the presence of a single feedback mirror, as shown in Fig. 9 [101]. Obviously, the patterns seen in these systems are strikingly similar to those observed in hydrodynamics.

One other optical system that displays complex behavior is a resonator that contains a phase-conjugate mirror. A phase-conjugate mirror has the peculiar property that it reflects a field whose amplitude is the complex conjugate of the incident field; in essence, it perfectly reverses the incoming wave front [45,47]. Hence, any transverse beam profile is an allowed mode of such a resonator. It is found that the light emitted by a phase conjugate resonator can display chaotic spatial–temporal dynamics, as shown in Fig. 10, which may be due to defect-mediated turbulence [102]. In addition, researchers have observed the transition from boundary-to-bulk dominated patterns [103] in a phase-conjugate resonator.

As I mentioned earlier, the study of optical turbulence is a rapidly emerging field in optical physics. We are just beginning to understand and characterize the behavior of these systems, and I expect that many new advances will be made in the next decade. We may even find wonderful new applications of these effects especially in the area of optical computers and pattern recognition.

VI. Controlling Optical Chaos

Most of the research on the dynamics of optical systems has involved the identification and characterization of systems that displayed deter-

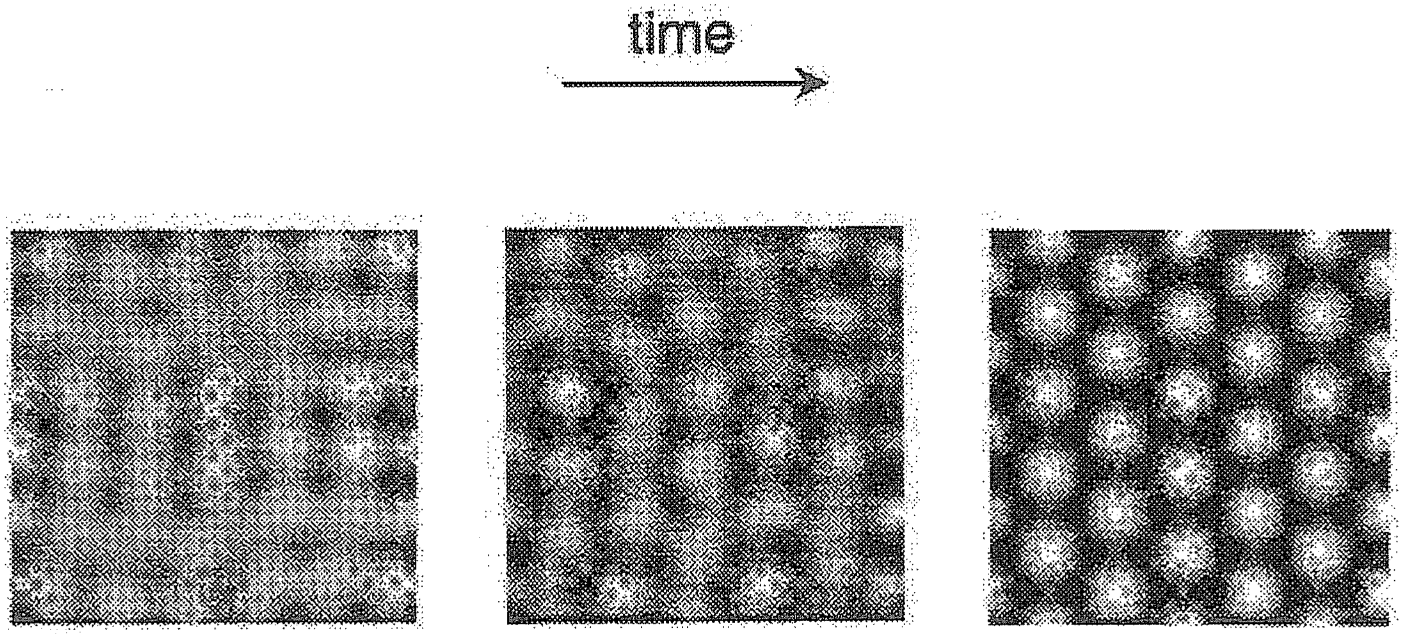

Figure 9. Formation of hexagonal spatial patterns in the interaction of a laser beam with a Kerr nonlinear medium in the presence of a single feedback mirror. Reprinted from W. J. Firth, Hexagon patterns and related phenomena in nonlinear optics, in *Chaos in Optics*, R. Roy, Ed., Proc. Soc. Photo-Opt. Instrum. Eng. **2039** (1993), pp. 290–303.

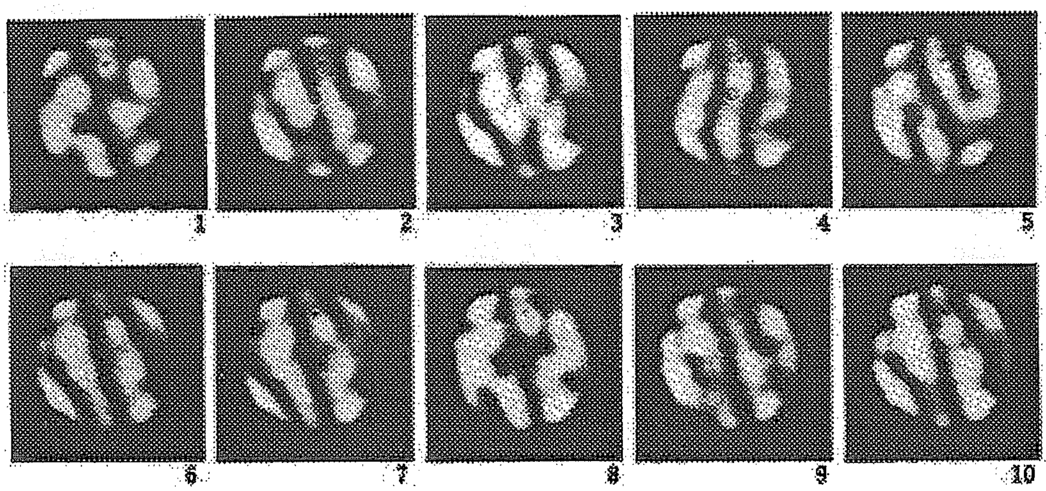

Figure 10. Temporal evolution of the transverse profile of the radiation emitted by a phase-conjugate resonator. Reprinted from S. R. Liu and G. Indebetouw, Periodic and chaotic spatiotemporal states in a phase-conjugate resonator using a photorefractive $BaTiO_3$ phase-conjugate mirror, J. Opt. Soc. Am. B **9**, 1507–1520 (1992).

ministic behavior. These studies resulted in an understanding of the mechanisms responsible for the instabilities and determined the domain of instabilities. This has been a great aid to optical engineers, because systems could be designed with an optimized domain of stability. However, designers have still been in the dark when it has been difficult or costly to redesign a system undergoing chaotic fluctuations.

Recently, several research groups [34–41] have offered a solution to this problem. They have found that it is possible (and quite easy) to stabilize chaotic laser systems. The reason these systems can be stabilized is that the strange attractor describing the dynamics of the system is arbitrarily close to an infinite number of unstable periodic orbits. If the system can be placed on one of the unstable periodic orbits, it will remain there until noise drives it off [31–33]. Hence, once the system is on an orbit, only minimal control must be applied to compensate for the noise in the system. The technique has been used to increase significantly the useful power generated by lasers, as shown in Fig. 11, where a tracking scheme was employed to follow changes in the system when the pump power was increased [35]. Other researchers are using the control technique to find the unstable periodic orbits and subsequently characterize the strange attractor in which they are embedded [36,37].

I suspect that in the very near future we will see a melding of the research on extended optical systems and controlling deterministic chaos. Several groups are pondering ways to control the spatial-temporal dynamics described in the previous section.

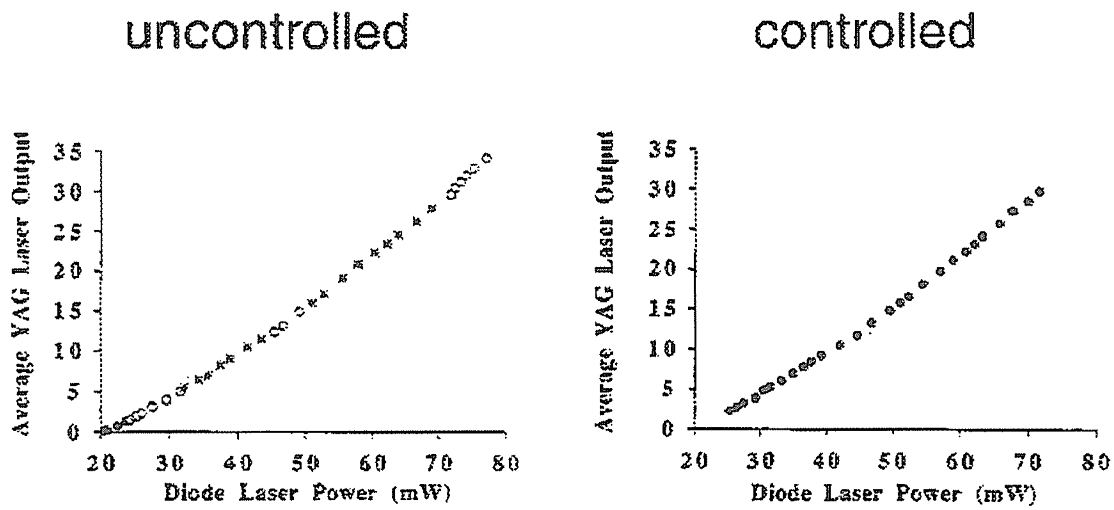

Figure 11. Average output power of an uncontrolled and controlled chaotic laser. The solid circles indicate stable operation of the laser. In the controlled laser the regime of stable operation was extended from about 20% above threshold to more than 300% above threshold. Reprinted from R. Roy, T. W. Murphy, Jr., T. D. Maier, Z. Gills, and E. R. Hunt, Dynamical control of a chaotic laser: Experimental stabilization of a globally coupled system, Phys. Rev. Lett. **68**, 1259–1262 (1992).

VII. The Quantum Nature of Optical Instabilities

Although it was not emphasized in the previous sections, the quantum mechanical nature of the radiation field is accentuated near the threshold for instabilities in some optical systems. For example, non-classical states of the radiation field can be generated by optical parametric oscillators operating just below threshold [104], spontaneous emission (due to quantum fluctuations) can obscure the appearance of deterministic instabilities [105], and the particle nature of the field can washout the threshold for bistability [106]. Currently, researchers are quantizing the models described in the previous sections to investigate how transverse effects modify the light–matter interaction at the quantum level [107].

VIII. Musings

I have attempted to give a flavor of our understanding of the dynamics of optical systems by giving a few examples from past and current research. Of course, these examples represent a very small fraction of the work in the field. I find that this is a very exciting period in the study of optical systems because we are becoming more aware of our inadequate understanding of complex systems as we learn more about the simple ones. I strongly urge the interested reader to consult the excellent textbooks and review articles on the subject.

Acknowledgments

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Note added in proof: Two excellent companion articles written by Lugiato [108] and Weiss [109] that review spatial-temporal structures in optical systems were inadvertently left out of the reference list in the original manuscript.

References

[1] R. J. Collins, D. F. Nelson, A. L. Schawlow, W. Bond, C. G. B. Garrett, and W. Kaiser, Coherence, narrowing, directionality, and relaxation oscillations in the light emission from ruby, Phys. Rev. Lett. **5**, 303–305 (1960).
[2] D. F. Nelson and W. S. Boyle, A continuously operating ruby optical laser, Appl. Opt. **1**, 181–183 (1962).
[3] C. M. Bowden, M. Ciftan, and H. R. Robl, Eds., *Optical Bistability* (Plenum, New York, 1981).
[4] C. M. Bowden, H. M. Gibbs, and S. L. McCall, *Optical Bistability II* (Plenum, New York, 1984).
[5] H. M. Gibbs, P. Mandel, N. Peyghambarian, and S. D. Smith, Eds., *Optical Bistability III* (Springer-Verlag, Berlin, 1986).
[6] W. Firth, N. Peyghambarian, and A. Tallet, Eds., *Optical Bistability IV*, J. Phys. (Paris) 49, Colloque C2, 483–488 (1988).
[7] L. A. Lugiato, Optical bistability, in *Progress in Optics*, E. Wolf, Ed. (North-Holland, Amsterdam, 1984), Vol. XXI, pp. 69–216.
[8] B. S. Wherrett and S. D. Smith, Eds., *Optical Bistability, Dynamical Nonlinearity and Photonic Logic* (The Royal Society, London, 1985).
[9] H. M. Gibbs, *Optical Bistability—Controlling Light with Light* (Academic, New York, 1985).
[10] R. G. Harrison and D. J. Biswas, Pulsating instabilities and chaos in lasers, Prog. Quantum Electron. **4**, 147–228 (1985).
[11] R. W. Boyd, M. G. Raymer, and L. M. Narducci, Eds., *Optical Instabilities* (Cambridge University Press, Cambridge, 1986).
[12] J. Chrostowski and N. B. Abraham, Eds., *Optical Chaos*, Proc. Soc. Photo-Opt. Instrum. Eng. **667** (1986).
[13] F. T. Arecchi and R. G. Harrison, Eds., *Instabilities and Chaos in Quantum Optics* (Springer-Verlag, Berlin, 1987).
[14] R. Graham and A. Wunderlin, Eds., *Lasers and Synergetics* (Springer-Verlag, Berlin, 1987).
[15] P. W. Milonni, J. R. Ackerhalt, and M.-L. Shih, *Chaos in Light–Matter Interactions* (World Scientific, Singapore, 1987).
[16] N. B. Abraham, F. T. Arecchi, and L. A. Lugiato, Eds., *Instabilities and Chaos in Quantum Optics II* (Plenum, New York, 1988).
[17] L. M. Narducci and N. B. Abraham, *Laser Physics and Laser Instabilities* (World Scientific, Singapore, 1988).
[18] N. B. Abraham, P. Mandel, and L. M. Narducci, Dynamical instabilities and pulsations in lasers, in *Progress in Optics*, E. Wolf, Ed. (North-Holland, Amsterdam, 1988), Vol. XXV, pp. 1–199.
[19] N. B. Abraham, Complexities and progress in studies of optical chaos, Opt. News **15**, 8–12 (1989).
[20] L. Pesquera and F. J. Bermejo, Eds., *Dynamics of Non-Linear Optical Systems* (World Scientific, Singapore, 1989).
[21] C. O. Weiss and R. Vilaseca, *Dynamics of Lasers* (VCH, Weinheim, 1991).
[22] F. Mitschke, Ed., "Special Issue on Chaos in Nonlinear Optics," Appl. Phys. B **55**, 103–138 (1992).
[23] N. B. Abraham, A. M. Albano, A. Passamante, P. E. Rapp, and R. Gilmore, Eds., Proceedings of the second workshop on measures of

complexity and chaos, Int. J. Bifurcation Chaos **3**, 485–779 (1993).

[24] R. Roy, Ed., *Chaos in Optics*, Proc. Soc. Photo-Opt. Instrum. Eng. **2039** (1993).

[25] R. G. Harrison and J. S. Uppal, Eds., *Nonlinear Dynamics and Spatial Complexity in Optical Systems* (Institute of Physics Publishing, Colchester, VT, 1993).

[26] N. B. Abraham and W. J. Firth, Overview of transverse effects in nonlinear-optical systems, J. Opt. Soc. Am. B **7**, 951–962 (1990).

[27] N. B. Abraham and W. J. Firth, Eds., Special issues on "Transverse effects in nonlinear-optical systems," J. Opt. Soc. Am. B **7**, 948–1157 and 1264–1373 (1990).

[28] R. Lang and K. Kobayashi, External optical feedback effects on semiconductor injection laser properties, IEEE J. Quantum Electron. **QE-16**, 347–355 (1980).

[29] K. Stubkjaer and M. B. Small, Feedback-induced noise in index-guided semiconductor lasers and its reduction by modulation, Electron. Lett. **19**, 388–389 (1983).

[30] A. Arimoto, M. Ojima, N. Chinone, A. Oishi, T. Gotoh, and N. Ohnuki, Optimum conditions for the high frequency noise reduction method in optical videodisc players, Appl. Opt. **25**, 1398–1403 (1986).

[31] E. Ott, C. Grebogi, and J. A. Yorke, Controlling chaos, Phys. Rev. Lett. **64**, 1196–1199 (1990).

[32] T. Shinbrot, C. Grebogi, E. Ott, and J. A. Yorke, Using small perturbations to control chaos, Nature **363**, 411–417 (1993).

[33] T. Shinbrot, Chaos: Unpredictable yet controllable?, Nonlin. Sci. Today **3**, 1–8 (1993).

[34] R. Roy, T. W. Murphy, Jr., T. D. Maier, Z. Gills, and E. R. Hunt, Dynamical control of a chaotic laser: Experimental stabilization of a globally coupled system, Phys. Rev. Lett. **68**, 1259–1262 (1992).

[35] Z. Gills, C. Iwata, R. Roy, I. B. Schwartz, and I. Triandaf, Tracking unstable steady states: Extending the stability regime of a multimode laser, Phys. Rev. Lett. **69**, 3169–3172 (1992).

[36] S. Bielawski, D. Derozier, and P. Glorieux, Experimental characterization of unstable periodic orbits by controlling chaos, Phys. Rev. A **47**, R2492–R2495 (1993).

[37] S. Bielawski, M. Bouazaoui, D. Derozier, and P. Glorieux, Stabilization and characterization of unstable steady states in a laser, Phys. Rev. A **47**, 3276–3279 (1993).

[38] G. R. Gray, A. T. Ryan, G. P. Agarwal, and E. C. Gage, Optical-feedback-induced chaos and its control in semiconductor lasers, in *Chaos in Optics*, R. Roy, Ed., Proc. Soc. Photo-Opt. Instrum. Eng. **2039** (1993), pp. 45–57.

[39] A. Gavrielides, V. Kovanis, and P. Alsing, Controlling chaos in semiconductor laser devices, in *Chaos in Optics*, R. Roy, Ed., Proc. Soc. Photo-Opt. Instrum. Eng. **2039** (1993), pp. 250–262.

[40] S. Bielawski, D. Derozier, and P. Glorieux, Controlling unstable periodic orbits by a delayed continuous feedback, to appear in Phys. Rev. E (1994).

[41] A. Gavrielides and V. Kovanis, Taming chaos in semiconductor lasers with optical feedback, preprint.

[42] N. Bloembergen, *Nonlinear Optics* (Benjamin/Cummings, Reading, MA, 1965).

[43] F. Zernike and J. E. Midwinder, *Applied Nonlinear Optics* (J Wiley, New York, 1973).

[44] H. Haken, *Light, Volume 3: Nonlinear Optics* (North-Holland, Amsterdam, 1982).

[45] R. A. Fisher, Ed., *Optical Phase Conjugation* (Academic, New York, 1983).

[46] Y. R. Shen, *The Principles of Nonlinear Optics* (J Wiley, New York, 1984).

[47] B. T. Zeldovich, N. F. Pipiletsky, and V. V. Shkunov, *Principles of Phase Conjugation* (Springer-Verlag, New York, 1985).

[48] M. Schubert and B. Wilhelmi, *Nonlinear Optics and Quantum Electronics* (J Wiley, New York, 1986).

[49] R. W. Boyd, *Nonlinear Optics* (Academic, Boston, 1992).

[50] A. C. Newell and J. V. Moloney, *Nonlinear Optics* (Addison-Wesley, Reading, MA, 1992).

[51] R. W. Boyd and G. P. Agrawal, Eds., *Contemporary Nonlinear Optics* (Academic, Boston, 1992).

[52] Y. R. Shen, Self-focusing: Experimental, Prog. Quantum Electron. **4**, 1–34 (1975).

[53] J. H. Marburger, Self-focusing: Theory, Prog. Quantum Electron. **4**, 35–110 (1975).

[54] H. M. Gibbs, B. Bolger, F. P. Mattar, M. C. Newstein, G. Forster, and P. E. Toschek, Coherent on-resonance self-focusing of optical pulses in absorbers, Phys. Rev. Lett. **37**, 1743–1746 (1975).

[55] L. A. Lugiato, R. J. Horowicz, G. Strini, and L. M. Narducci, Instabilities in passive and active systems with a Gaussian transverse intensity profile, Phys. Rev. A **30**, 1366–1376 (1984).

[56] L. A. Lugiato and M. Milani, Effects of Gaussian-beam averaging on laser instabilities, J. Opt. Soc. Am. B **2**, 15–17 (1985).

[57] A. L. Schawlow and C. H. Townes, Infrared and optical masers, Phys. Rev. **112**, 1940–1949 (1958); see also: M. Born and E. Wolf, *Principle of Optics*, 6th ed. (Pergamon, Oxford, 1980), Sec. 8.3.3.

[58] A. Z. Grasiuk and A. N. Oraevsky, The dynamics of quantum oscillators, in *Quantum Electronics and Coherent Light*, P. A. Miles, Ed. (Academic, New York, 1964), pp. 192–198.

[59] V. V. Koroblin and A. V. Upenskiy, On the theory of the pulsations in the output of the ruby laser, Zh. Eksp. Teor. Fiz. **45**, 1003–1008 (1963) [Sov. Phys. JETP **18**, 693–697 (1964)].

[60] H. Haken, Theory of intensity and phase fluctuations of a homogeneously broadened laser, Z. Phys. **190**, 327–356 (1966).

[61] H. Risken and K. Nummendal, Self-pulsing in lasers, J. Appl. Phys. **39**, 4662–4672 (1968).

[62] H. Haken, Analogy between higher instabilities in fluids and lasers, Phys. Lett. **53A**, 77–78 (1975).

[63] E. N. Lorenz, Deterministic nonperiodic flow, J. Atmos. Sci. **20**, 130 (1963).

[64] C. O. Weiss and W. Klische, On observability of Lorenz instabilities in lasers, Opt. Commun. **51**, 47–48 (1984).

[65] C. O. Weiss, N. B. Abraham, and U. Hübner, Homoclinic and heteroclinic chaos in a single-mode laser, Phys. Rev. Lett. **61**, 1587–1590 (1988).

[66] U. Hübner, N. B. Abraham, and C. O. Weiss, Dimensions and entropies of chaotic intensity pulsations in a single-mode far-infrared NH_3 laser, Phys. Rev. A **40**, 6354–6365 (1989).

[67] E. Roldan, G. J. de Valcarcel, R. Vilaseca, and P. Mandel, Single-mode-laser phase dynamics, Phys. Rev. A **48**, 591–598 (1993).

[68] B. Meziane, Instability hierarchies in self-pulsing lasers, Phys. Rev. A **48**, 2346–2360 (1993).

[69] H. J. Carmichael, Overview of the theory of optical bistability, in *Optical Instabilities*, R. W. Boyd, M. G. Raymer, and L. M. Narducci, Eds. (Cambridge University Press, Cambridge, 1986), pp. 111–128.

[70] K. Ikeda, H. Daido, and O. Akimoto, Optical turbulence: chaotic behavior of transmitted light from a ring cavity, Phys. Rev. Lett. **45**, 709–712 (1980).

[71] H. M. Gibbs, F. A. Hopf, D. L. Kaplan, and R. L. Shoemaker, Observation of chaos in optical bistability, Phys. Rev. Lett. **46**, 474–477 (1981).

[72] H. Nakatsuka, S. Asaka, H. Itoh, K. Ikeda, and M. Matsuoka, Observation of bifurcation to chaos in an all-optical bistable system, Phys. Rev. Lett. **50**, 109–112 (1983).

[73] A. T. Rosenberger, L. A. Orozco, and H. J. Kimble, Observation of absorptive bistability with two-level atoms in a ring cavity, Phys. Rev. A **28**, 2569–2572 (1983).

[74] L. A. Orozco, H. J. Kimble, and A. T. Rosenberger, Quantitative test of the single-mode theory of optical bistability, Opt. Commun. **62**, 54–60 (1987).

[75] L. A. Orozco, A. T. Rosenberger, and H. J. Kimble, Optical bistability in the mixed absorptive-dispersive regime with two-state atoms, Phys. Rev. A **36**, 3248–3252 (1987).

[76] Y. Silberberg and I. Bar-Joseph, Instabilities, self-oscillation and chaos in a simple nonlinear optical interaction, Phys. Rev. Lett. **48**, 1541–1543 (1982).

[77] Y. Silberberg and I. Bar-Joseph, Optical Instabilities in a nonlinear Kerr medium, J. Opt. Soc. Am. B **1**, 662–670 (1984).

[78] G. Khitrova, J. F. Valley, and H. M. Gibbs, Gain-feedback approach to optical instabilities in sodium vapor, Phys. Rev. Lett. **60**, 1126–1129 (1988).

[79] W. J. Firth and C. Paré, Transverse modulational instabilities for counterpropagating beams in Kerr media, Opt. Lett. **13**, 1096–1098 (1988).

[80] W. J. Firth, A. Fitzgerald, and C. Paré, Transverse instabilities due to counterpropagation in Kerr media, J. Opt. Soc. Am. B **7**, 1087–1097 (1990).

[81] A. L. Gaeta, R. W. Boyd, J. R. Ackerhalt, and P. W. Milonni, Instabilities and chaos in the polarizations of counterpropagating light fields, Phys. Rev. Lett. **58** 2432–2435 (1987).

[82] G. Grynberg, Mirrorless four-wave mixing oscillation in atomic vapors, Opt. Commun. **66**, 321–324 (1988).

[83] G. Grynberg, E. LeBihan, P. Verkerk, P. Simoneau, J. R. R. Leite, D. Bloch, S. LeBoiteux, and M. Ducloy, Observation of instabilities due to

mirrorless four-wave mixing oscillation in sodium, Opt. Commun. **67**, 363–366 (1988).
[84] G. Grynberg and J. Paye, Spatial instability for a standing wave in a nonlinear medium, Europhys. Lett. **8**, 29–33 (1989).
[85] D. J. Gauthier, M. S. Malcuit, and R. W. Boyd, Polarization instabilities of counterpropagating laser beams in sodium vapor, Phys. Rev. Lett. **61**, 1827–1830 (1988).
[86] D. J. Gauthier, M. S. Malcuit, A. L. Gaeta, and R. W. Boyd, Polarization bistability of counterpropagating laser beams, Phys. Rev. Lett. **64**, 1721–1724 (1990).
[87] P. Narum, A. L. Gaeta, M. D. Skeldon, and R. W. Boyd, Instabilities of laser beams counterpropagating through a Brillouin-active medium, J. Opt. Soc. Am. B **5**, 623–628 (1988).
[88] A. L. Gaeta, M. D. Skeldon, R. W. Boyd, and P. Narum, Observation of instabilities of laser beams counterpropagating through a Brillouin medium, J. Opt. Soc. Am. B **5**, 1709–1712 (1989).
[89] A. L. Gaeta and R. W. Boyd, Transverse instabilities in the polarizations and intensities of counterpropagating light waves, Phys. Rev. A **48**, 1610–1624 (1993).
[90] M. Saffman, D. Montgomery, A. A. Zozulya, K. Kuroda, and D. Z. Anderson, Transverse instabilities of counterpropagating waves in photorefractive media, Phys. Rev. A **48**, 3209–3215 (1993).
[91] J. W. Grantham, H. M. Gibbs, G. Khitrova, J. F. Valley, and X. Jiajin, Kaleidoscopic spatial instability: Bifurcations of optical transverse solitary waves, Phys. Rev. Lett. **66**, 1422–1425 (1991).
[92] C. T. Law and G. A. Swartzlander, Jr., Optical vortex solitons and the stability of dark soliton stripes, Opt. Lett. **18**, 586–588 (1993).
[93] F. T. Arecchi, Space-time complexity in nonlinear optics, Physica D **51**, 450–464 (1991).
[94] K. Staliunas, Laser Ginzburg–Landau equation and laser hydrodynamics, Phys. Rev. A **48**, 1573–1581 (1993).
[95] P. La Penna and G. Giusfredi, Spatiotemporal instabilities in a Fabry–Perot resonator filled with sodium vapor, Phys. Rev. A **48**, 2299–2314 (1993).
[96] M. Tlidi, M. Georgiou, and P. Mandel, Transverse patterns in nascent optical bistability, Phys. Rev. A **48**, 4605–4609 (1993).
[97] H. Lin and N. B. Abraham, Low-frequency pulsations in a He-Xe laser with several transverse modes: Experimental and numerical studies, Phys. Rev. A **49**, 2076–2086 (1994).
[98] Q. Feng, J. V. Moloney, and A. C. Newell, Amplitude instabilities of transverse traveling waves in lasers, Phys. Rev. Lett. **71**, 1705–1708 (1993).
[99] M. Brambilla, D. Camesasca, A. Gatti, L. A. Lugiato, and G. L. Oppo, Spontaneous spatial structures and spatial squeezing in optical parametric oscillators, in *Chaos in Optics*, R. Roy, Ed., Proc. Soc. Photo-Opt. Instrum. Eng. **2039** (1993), pp. 282–289.
[100] G.-L. Oppo, M. Brambilla and L. Lugiato, Formation and evolution of roll patterns in optical parametric oscillators, Phys. Rev. A **49**, 2028–2032 (1994).
[101] W. J. Firth, Hexagon patterns and related phenomena in nonlinear optics, in *Chaos in Optics*, R. Roy, Ed., Proc. Soc. Photo-Opt. Instrum. Eng. **2039** (1993), pp. 290–303.
[102] S. R. Liu and G. Indebetouw, Periodic and chaotic spatiotemporal states in a phase-conjugate resonator using a photorefractive $BaTiO_3$ phase-conjugate mirror, J. Opt. Soc. Am. B **9**, 1507–1520 (1992).
[103] F. T. Arecchi, S. Boccaletti, P. L. Ramazza, and S. Residori, Transition from boundary- to bulk-controlled regimes in optical pattern formation, Phys. Rev. Lett. **70**, 2277–2280 (1993).
[104] L.-A. Wu, H. J. Kimble, J. L. Hall, and H. Wu, Generation of squeezed states by parametric down conversion, Phys. Rev. Lett. **57**, 2520–2523 (1986).
[105] M. Dammig, G. Zinner, F. Mitschke, and H. Welling, Stimulated Brillouin scattering in fibers with and without external feedback, Phys. Rev. A **48**, 3301–3309 (1993).
[106] G. Rempe, R. J. Thompson, R. J. Brecha, W. D. Lee, and H. J. Kimble, Optical bistability and photon statistics in cavity quantum electrodynamics, Phys. Rev. Lett. **67**, 1727–1730 (1991).
[107] L. A. Lugiato and A. Gatti, Spatial structure of a squeezed vacuum, Phys. Rev. Lett. **70**, 3868–3871 (1993).
[108] L. A. Lugiato, Spatio-temporal structures. Part I, Phys. Rep. **219**, 293–310 (1992).
[109] C. O. Weiss, Spatio-temporal structures. Part II. Vorticies and defects in lasers, Phys. Rep. **219**, 311–338 (1992).